



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - DEC 2023

UE18CS353 - Object Oriented Analysis and Design with Software Engineering

Total Marks : 100.0

1.a. With Diagram, Explain the software development life cycle(SDLC)
(8.0 Marks)

1.b. List 8 activities that are performed during the feasibility study of the requirement analysis.
(8.0 Marks)

1.c. Explain the advantages of waterfall model
(4.0 Marks)

2.a. Identify the at least 6 classes from the below case study:

Design the software to support a computerized banking network including both human cashiers and automatic teller machines (ATMs) to be shared by a consortium of banks. Each bank provides its computer to maintain its own accounts and process transactions against them. Cashier stations are owned by individual banks and communicate directly with their own bank's computers. Human cashiers enter account and transaction data. Automatic teller machines communicate with a central computer that clears transactions with the appropriate banks. An automatic teller machine accepts a cash card, interacts with the user, communicates with the central system to carry out the transaction, dispenses cash, and prints receipts. The system requires appropriate recordkeeping and security provisions. The system must handle concurrent access to the same account correctly.

The banks will provide their software for their computers; you are to design the software for the ATMs and the network. The cost of the shared system will be apportioned to the banks according to the number of customers with cash cards. (6.0 Marks)

2.b. Draw Component diagram for ATM Components (8.0 Marks)

2.c. Draw a sequence diagram for a phone calling application (6.0 Marks)

3.a. Explain the single responsibility principle

(6.0 Marks)

3.b. Explain briefly the code review and code inspection

(8.0 Marks)

3.c. Explain Software Configuration Management and its benefits

(6.0 Marks)

4.a. Explain briefly about the 4 pillars of DevOps

(8.0 Marks)

4.b. With Diagram explain the levels of testing

(8.0 Marks)

4.c. Write a note on Black box & White box testing

(4.0 Marks)

5.a. Explain the key issues in the software maintenance

(8.0 Marks)

5.b. With Diagram explain the process of test execution

(6.0 Marks)

5.c. Differentiate between VERIFICATION & VALIDATION

(6.0 Marks)